

```

import pandas as pd

df = pd.read_csv("Cricket_data.csv")

df.drop_duplicates(inplace=True)

num_cols = df.select_dtypes(include=["int64", "float64"]).columns
df[num_cols] = df[num_cols].fillna(df[num_cols].median())

cat_cols = df.select_dtypes(include=["object"]).columns
df[cat_cols] = df[cat_cols].fillna(df[cat_cols].mode().iloc[0])

df["start_date"] = pd.to_datetime(df["start_date"], errors="coerce")
df["end_date"] = pd.to_datetime(df["end_date"], errors="coerce")

df.columns = (
    df.columns
    .str.strip()
    .str.lower()
    .str.replace(" ", "_")
)

print(df.info())
print(df.isnull().sum())

df.to_csv("Cricket_data.csv", index=False)

```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1032 entries, 0 to 1031
Data columns (total 43 columns):
#   Column                Non-Null Count  Dtype
---  -
0   season                1032 non-null   float64
1   id                    1032 non-null   int64
2   name                  1032 non-null   object
3   short_name            1032 non-null   object
4   description            1032 non-null   object
5   home_team             1032 non-null   object
6   away_team             1032 non-null   object
7   toss_won              1032 non-null   object
8   decision              1032 non-null   object
9   1st_inning_score      1032 non-null   object
10  2nd_inning_score      1032 non-null   object
11  winner                1032 non-null   object
12  result                1032 non-null   object
13  start_date            1032 non-null   datetime64[ns, UTC]
14  end_date              1032 non-null   datetime64[ns, UTC]
15  venue_id              1032 non-null   int64
16  venue_name            1032 non-null   object
17  home_captain          1032 non-null   object

```

18	away_captain	1032	non-null	object
19	pom	1032	non-null	object
20	points	1032	non-null	object
21	super_over	1032	non-null	bool
22	home_overs	1032	non-null	float64
23	home_runs	1032	non-null	float64
24	home_wickets	1032	non-null	float64
25	home_boundaries	1032	non-null	float64
26	away_overs	1032	non-null	float64
27	away_runs	1032	non-null	float64
28	away_wickets	1032	non-null	float64
29	away_boundaries	1032	non-null	float64
30	highlights	1032	non-null	object
31	home_key_batsman	1032	non-null	object
32	home_key_bowler	1032	non-null	object
33	home_playx1	1032	non-null	object
34	away_playx1	1032	non-null	object
35	away_key_batsman	1032	non-null	object
36	away_key_bowler	1032	non-null	object
37	match_days	1032	non-null	object
38	umpire1	1032	non-null	object
39	umpire2	1032	non-null	object
40	tv_umpire	1032	non-null	object
41	referee	1032	non-null	object
42	reserve_umpire	1032	non-null	object

dtypes: bool(1), datetime64[ns, UTC](2), float64(9), int64(2), object(29)
 memory usage: 339.8+ KB
 None

season	0
id	0
name	0
short_name	0
description	0
home_team	0
away_team	0
toss_won	0
decision	0
1st_inning_score	0
2nd_inning_score	0
winner	0
result	0
start_date	0
end_date	0
venue_id	0
venue_name	0
home_captain	0
away_captain	0
pom	0

```

points          0
super_over      0
home_overs      0
home_runs       0
home_wickets    0
home_boundaries 0
away_overs      0
away_runs       0
away_wickets    0
away_boundaries 0
highlights      0
home_key_batsman 0
home_key_bowler 0
home_playx1     0
away_playx1     0
away_key_batsman 0
away_key_bowler 0
match_days      0
umpire1         0
umpire2         0
tv_umpire       0
referee         0
reserve_umpire  0
dtype: int64

```

```
import pandas as pd
```

```
df = pd.read_csv("Cricket_data.csv")
```

```
df["season"] = df["season"].fillna(df["season"].median())
```

```
df["toss_won"] = df["toss_won"].fillna(df["toss_won"].mode()[0])
```

```
df["decision"] = df["decision"].fillna(df["decision"].mode()[0])
```

```
df["home_captain"] =
df["home_captain"].fillna(df["home_captain"].mode()[0])
```

```
df["away_captain"] =
df["away_captain"].fillna(df["away_captain"].mode()[0])
```

```
df["pom"] = df["pom"].fillna(df["pom"].mode()[0])
```

```
df.drop_duplicates(inplace=True)
```

```
print(df.isnull().sum())
```

```

season          0
id              0
name            0
short_name      0

```

```
description      0
home_team        0
away_team        0
toss_won         0
decision         0
1st_inning_score 0
2nd_inning_score 0
winner           0
result           0
start_date       0
end_date         0
venue_id         0
venue_name       0
home_captain     0
away_captain     0
pom              0
points           0
super_over       0
home_overs       0
home_runs        0
home_wickets     0
home_boundaries  0
away_overs       0
away_runs        0
away_wickets     0
away_boundaries  0
highlights       0
home_key_batsman 0
home_key_bowler  0
home_playx1      0
away_playx1      0
away_key_batsman 0
away_key_bowler  0
match_days       0
umpire1          0
umpire2          0
tv_umpire        0
referee          0
reserve_umpire   0
dtype: int64
```

```
import pandas as pd

df = pd.read_csv("Cricket_data.csv")

print(df.head())

print(df.info())

print(df.isnull().sum())
```

season	id		name	
short_name \				
0 2015.0	1370350	Chennai Super Kings v Gujarat Titans		CSK
v GT				
1 2015.0	1370351	Lucknow Super Giants v Mumbai Indians		LSG
v MI				
2 2015.0	1370352	TBC v TBC		TBC
v TBC				
3 2015.0	1370353	TBC v TBC		TBC
v TBC				
4 2023.0	1359544	Royal Challengers Bangalore v Gujarat Titans		RCB
v GT				
		description	home_team	
away_team \				
0 Qualifier 1 (N), Indian Premier League at Chen...			CSK	
GT				
1 Eliminator (N), Indian Premier League at Chenn...			LSG	
MI				
2 Qualifier 2 (N), Indian Premier League at Ahme...			TBA	
TBA				
3 Final (N), Indian Premier League at Ahmedabad,...			TBA	
TBA				
4 70th Match (N), Indian Premier League at Benga...			RCB	
GT				
toss_won	decision	1st_inning_score	...	\
0 MI	BOWL FIRST	187/5	...	
1 MI	BOWL FIRST	187/5	...	
2 MI	BOWL FIRST	187/5	...	
3 MI	BOWL FIRST	187/5	...	
4 GT	BOWL FIRST	197/5	...	
		home_playx1	\	
0 Adam Gilchrist (WK),VVS Laxman (BT),Herschelle...				
1 Adam Gilchrist (WK),VVS Laxman (BT),Herschelle...				
2 Adam Gilchrist (WK),VVS Laxman (BT),Herschelle...				
3 Adam Gilchrist (WK),VVS Laxman (BT),Herschelle...				
4 Virat Kohli (UKN),Faf du Plessis (UKN),Glenn M...				
		away_playx1	\	
0 Abhishek Sharma (AR),Kane Williamson (UKN),Rah...				
1 Abhishek Sharma (AR),Kane Williamson (UKN),Rah...				
2 Abhishek Sharma (AR),Kane Williamson (UKN),Rah...				
3 Abhishek Sharma (AR),Kane Williamson (UKN),Rah...				
4 Wriddhiman Saha (WK),Shubman Gill (UKN),Vijay ...				
		away_key_batsman	away_key_bowler	\
0 Chris Gayle,Virat Kohli	Harshal Patel,Yuzvendra Chahal			
1 Chris Gayle,Virat Kohli	Harshal Patel,Yuzvendra Chahal			

2	Chris Gayle, Virat Kohli	Harshal Patel, Yuzvendra Chahal
3	Chris Gayle, Virat Kohli	Harshal Patel, Yuzvendra Chahal
4	Shubman Gill, Vijay Shankar	Noor Ahmad, Rashid Khan

	match_days	umpire1	\
0	8 October 2021 - night match (20-over match)	Kumar Dharmasena	
1	8 October 2021 - night match (20-over match)	Kumar Dharmasena	
2	8 October 2021 - night match (20-over match)	Kumar Dharmasena	
3	8 October 2021 - night match (20-over match)	Kumar Dharmasena	
4	21 May 2023 - night match (20-over match)	Nitin Menon	

	umpire2	tv_umpire	referee	
	reserve_umpire			
0	Sundaram Ravi	Anil Chaudhary	Javagal Srinath	Krishnamachari Srinivasan
1	Sundaram Ravi	Anil Chaudhary	Javagal Srinath	Krishnamachari Srinivasan
2	Sundaram Ravi	Anil Chaudhary	Javagal Srinath	Krishnamachari Srinivasan
3	Sundaram Ravi	Anil Chaudhary	Javagal Srinath	Krishnamachari Srinivasan
4	Virender Sharma	Tapan Sharma	Javagal Srinath	VM Dhokre

[5 rows x 43 columns]

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 1032 entries, 0 to 1031

Data columns (total 43 columns):

#	Column	Non-Null Count	Dtype
0	season	1032 non-null	float64
1	id	1032 non-null	int64
2	name	1032 non-null	object
3	short_name	1032 non-null	object
4	description	1032 non-null	object
5	home_team	1032 non-null	object
6	away_team	1032 non-null	object
7	toss_won	1032 non-null	object
8	decision	1032 non-null	object
9	1st_inning_score	1032 non-null	object
10	2nd_inning_score	1032 non-null	object
11	winner	1032 non-null	object
12	result	1032 non-null	object
13	start_date	1032 non-null	object
14	end_date	1032 non-null	object
15	venue_id	1032 non-null	int64
16	venue_name	1032 non-null	object
17	home_captain	1032 non-null	object
18	away_captain	1032 non-null	object
19	pom	1032 non-null	object

20	points	1032	non-null	object
21	super_over	1032	non-null	bool
22	home_overs	1032	non-null	float64
23	home_runs	1032	non-null	float64
24	home_wickets	1032	non-null	float64
25	home_boundaries	1032	non-null	float64
26	away_overs	1032	non-null	float64
27	away_runs	1032	non-null	float64
28	away_wickets	1032	non-null	float64
29	away_boundaries	1032	non-null	float64
30	highlights	1032	non-null	object
31	home_key_batsman	1032	non-null	object
32	home_key_bowler	1032	non-null	object
33	home_playx1	1032	non-null	object
34	away_playx1	1032	non-null	object
35	away_key_batsman	1032	non-null	object
36	away_key_bowler	1032	non-null	object
37	match_days	1032	non-null	object
38	umpire1	1032	non-null	object
39	umpire2	1032	non-null	object
40	tv_umpire	1032	non-null	object
41	referee	1032	non-null	object
42	reserve_umpire	1032	non-null	object

dtypes: bool(1), float64(9), int64(2), object(31)

memory usage: 339.8+ KB

None

season	0
id	0
name	0
short_name	0
description	0
home_team	0
away_team	0
toss_won	0
decision	0
1st_inning_score	0
2nd_inning_score	0
winner	0
result	0
start_date	0
end_date	0
venue_id	0
venue_name	0
home_captain	0
away_captain	0
pom	0
points	0
super_over	0
home_overs	0
home_runs	0

```
home_wickets      0
home_boundaries    0
away_overs         0
away_runs         0
away_wickets      0
away_boundaries    0
highlights        0
home_key_batsman   0
home_key_bowler    0
home_playx1       0
away_playx1       0
away_key_batsman   0
away_key_bowler    0
match_days        0
umpire1           0
umpire2           0
tv_umpire         0
referee           0
reserve_umpire     0
dtype: int64
```

```
df.drop_duplicates(inplace=True)
```

```
df["season"] = df["season"].fillna(df["season"].median())
```

```
df["toss_won"] = df["toss_won"].fillna(df["toss_won"].mode()[0])
```

```
df["decision"] = df["decision"].fillna(df["decision"].mode()[0])
```

```
df["home_captain"] =
df["home_captain"].fillna(df["home_captain"].mode()[0])
```

```
df["away_captain"] =
df["away_captain"].fillna(df["away_captain"].mode()[0])
```

```
df["pom"] = df["pom"].fillna(df["pom"].mode()[0])
```

```
df["home_runs"] = df["home_runs"].fillna(df["home_runs"].median())
```

```
df["away_runs"] = df["away_runs"].fillna(df["away_runs"].median())
```

```
df["home_wickets"] =
df["home_wickets"].fillna(df["home_wickets"].median())
```

```
df["away_wickets"] =
df["away_wickets"].fillna(df["away_wickets"].median())
```

```
print(df.isnull().sum())
```

```
season      0
id          0
name        0
```



```

short_name      0
description     0
home_team      0
away_team      0
toss_won       0
decision       0
1st_inning_score 0
2nd_inning_score 0
winner         0
result         0
start_date     0
end_date       0
venue_id       0
venue_name     0
home_captain   0
away_captain   0
pom           0
points         0
super_over     0
home_overs     0
home_runs      0
home_wickets   0
home_boundaries 0
away_overs     0
away_runs      0
away_wickets   0
away_boundaries 0
highlights     0
home_key_batsman 0
home_key_bowler 0
home_playx1    0
away_playx1    0
away_key_batsman 0
away_key_bowler 0
match_days     0
umpire1        0
umpire2        0
tv_umpire      0
referee        0
reserve_umpire 0
dtype: int64

df.to_csv("Cricket_data_cleaned.csv", index=False)

for col in df.columns:
    print(col)

season
id
name

```

```
short_name
description
home_team
away_team
toss_won
decision
1st_inning_score
2nd_inning_score
winner
result
start_date
end_date
venue_id
venue_name
home_captain
away_captain
pom
points
super_over
home_overs
home_runs
home_wickets
home_boundaries
away_overs
away_runs
away_wickets
away_boundaries
highlights
home_key_batsman
home_key_bowler
home_playx1
away_playx1
away_key_batsman
away_key_bowler
match_days
umpire1
umpire2
tv_umpire
referee
reserve_umpire
```

```
print(df["name"].isnull().sum())
```

```
0
```

```
print(df["name"])
```

```
0          Chennai Super Kings v Gujarat Titans
1    Lucknow Super Giants v Mumbai Indians
2                      TBC v TBC
```

```

3                                     TBC v TBC
4      Royal Challengers Bangalore v Gujarat Titans
...
1027    Royal Challengers Bangalore v Mumbai Indians
1028                Kings XI Punjab v Rajasthan Royals
1029                Delhi Daredevils v Rajasthan Royals
1030                Chennai Super Kings v Kings XI Punjab
1031                Chennai Super Kings v Rajasthan Royals
Name: name, Length: 1032, dtype: object

```

```
df["name"] = df["name"].fillna(df["name"].mode()[0])
```

```
name_list = df["name"].unique().tolist()
print(name_list)
```

```

['Chennai Super Kings v Gujarat Titans', 'Lucknow Super Giants v
Mumbai Indians', 'TBC v TBC', 'Royal Challengers Bangalore v Gujarat
Titans', 'Mumbai Indians v Sunrisers Hyderabad', 'Kolkata Knight
Riders v Lucknow Super Giants', 'Delhi Capitals v Chennai Super
Kings', 'Punjab Kings v Rajasthan Royals', 'Sunrisers Hyderabad v
Royal Challengers Bangalore', 'Punjab Kings v Delhi Capitals',
'Gujarat Titans v Sunrisers Hyderabad', 'Chennai Super Kings v Kolkata
Knight Riders', 'Rajasthan Royals v Royal Challengers Bangalore',
'Delhi Capitals v Punjab Kings', 'Sunrisers Hyderabad v Lucknow Super
Giants', 'Mumbai Indians v Gujarat Titans', 'Kolkata Knight Riders v
Rajasthan Royals', 'Chennai Super Kings v Delhi Capitals', 'Mumbai
Indians v Royal Challengers Bangalore', 'Kolkata Knight Riders v
Punjab Kings', 'Rajasthan Royals v Sunrisers Hyderabad', 'Gujarat
Titans v Lucknow Super Giants', 'Delhi Capitals v Royal Challengers
Bangalore', 'Chennai Super Kings v Mumbai Indians', 'Rajasthan Royals
v Gujarat Titans', 'Sunrisers Hyderabad v Kolkata Knight Riders',
'Punjab Kings v Mumbai Indians', 'Lucknow Super Giants v Chennai Super
Kings', 'Gujarat Titans v Delhi Capitals', 'Lucknow Super Giants v
Royal Challengers Bangalore', 'Mumbai Indians v Rajasthan Royals',
'Chennai Super Kings v Punjab Kings', 'Delhi Capitals v Sunrisers
Hyderabad', 'Kolkata Knight Riders v Gujarat Titans', 'Punjab Kings v
Lucknow Super Giants', 'Rajasthan Royals v Chennai Super Kings',
'Royal Challengers Bangalore v Kolkata Knight Riders', 'Gujarat Titans
v Mumbai Indians', 'Sunrisers Hyderabad v Delhi Capitals', 'Kolkata
Knight Riders v Chennai Super Kings', 'Royal Challengers Bangalore v
Rajasthan Royals', 'Mumbai Indians v Punjab Kings', 'Lucknow Super
Giants v Gujarat Titans', 'Chennai Super Kings v Sunrisers Hyderabad',
'Delhi Capitals v Kolkata Knight Riders', 'Punjab Kings v Royal
Challengers Bangalore', 'Rajasthan Royals v Lucknow Super Giants',
'Sunrisers Hyderabad v Mumbai Indians', 'Royal Challengers Bangalore v
Chennai Super Kings', 'Gujarat Titans v Rajasthan Royals', 'Mumbai
Indians v Kolkata Knight Riders', 'Lucknow Super Giants v Punjab
Kings', 'Royal Challengers Bangalore v Delhi Capitals', 'Kolkata
Knight Riders v Sunrisers Hyderabad', 'Punjab Kings v Gujarat Titans',
'Chennai Super Kings v Rajasthan Royals', 'Delhi Capitals v Mumbai

```

Indians', 'Royal Challengers Bangalore v Lucknow Super Giants', 'Sunrisers Hyderabad v Punjab Kings', 'Gujarat Titans v Kolkata Knight Riders', 'Mumbai Indians v Chennai Super Kings', 'Rajasthan Royals v Delhi Capitals', 'Lucknow Super Giants v Sunrisers Hyderabad', 'Kolkata Knight Riders v Royal Challengers Bangalore', 'Rajasthan Royals v Punjab Kings', 'Delhi Capitals v Gujarat Titans', 'Chennai Super Kings v Lucknow Super Giants', 'Royal Challengers Bangalore v Mumbai Indians', 'Sunrisers Hyderabad v Rajasthan Royals', 'Lucknow Super Giants v Delhi Capitals', 'Punjab Kings v Kolkata Knight Riders', 'Gujarat Titans v Chennai Super Kings', 'Kolkata Knight Riders v Mumbai Indians', 'Kolkata Knight Riders v Delhi Capitals', 'Sunrisers Hyderabad v Gujarat Titans', 'Chennai Super Kings v Royal Challengers Bangalore', 'Mumbai Indians v Lucknow Super Giants', 'Punjab Kings v Sunrisers Hyderabad', 'Rajasthan Royals v Kolkata Knight Riders', 'Delhi Capitals v Rajasthan Royals', 'Royal Challengers Bangalore v Sunrisers Hyderabad', 'Punjab Kings v Chennai Super Kings', 'Gujarat Titans v Royal Challengers Bangalore', 'Rajasthan Royals v Mumbai Indians', 'Delhi Capitals v Lucknow Super Giants', 'Sunrisers Hyderabad v Chennai Super Kings', 'Gujarat Titans v Punjab Kings', 'Lucknow Super Giants v Kolkata Knight Riders', 'Royal Challengers Bangalore v Punjab Kings', 'Lucknow Super Giants v Rajasthan Royals', 'Mumbai Indians v Delhi Capitals', 'Delhi Capitals v Kings XI Punjab', 'Kings XI Punjab v Royal Challengers Bangalore', 'Rajasthan Royals v Kings XI Punjab', 'Kings XI Punjab v Mumbai Indians', 'Kings XI Punjab v Chennai Super Kings', 'Sunrisers Hyderabad v Kings XI Punjab', 'Kings XI Punjab v Kolkata Knight Riders', 'Royal Challengers Bangalore v Kings XI Punjab', 'Mumbai Indians v Kings XI Punjab', 'Kings XI Punjab v Delhi Capitals', 'Kings XI Punjab v Sunrisers Hyderabad', 'Kolkata Knight Riders v Kings XI Punjab', 'Kings XI Punjab v Rajasthan Royals', 'Chennai Super Kings v Kings XI Punjab', 'Kings XI Punjab v Delhi Daredevils', 'Rajasthan Royals v Delhi Daredevils', 'Mumbai Indians v Delhi Daredevils', 'Kolkata Knight Riders v Delhi Daredevils', 'Royal Challengers Bangalore v Delhi Daredevils', 'Delhi Daredevils v Kings XI Punjab', 'Delhi Daredevils v Kolkata Knight Riders', 'Chennai Super Kings v Delhi Daredevils', 'Delhi Daredevils v Rajasthan Royals', 'Sunrisers Hyderabad v Delhi Daredevils', 'Delhi Daredevils v Sunrisers Hyderabad', 'Delhi Daredevils v Royal Challengers Bangalore', 'Delhi Daredevils v Chennai Super Kings', 'Delhi Daredevils v Mumbai Indians', 'Rising Pune Supergiant v Mumbai Indians', 'Gujarat Lions v Kolkata Knight Riders', 'Kings XI Punjab v Rising Pune Supergiant', 'Sunrisers Hyderabad v Gujarat Lions', 'Rising Pune Supergiant v Delhi Daredevils', 'Gujarat Lions v Rising Pune Supergiant', 'Mumbai Indians v Gujarat Lions', 'Royal Challengers Bangalore v Rising Pune Supergiant', 'Gujarat Lions v Royal Challengers Bangalore', 'Kolkata Knight Riders v Gujarat Lions', 'Rising Pune Supergiant v Sunrisers Hyderabad', 'Gujarat Lions v Kings XI Punjab', 'Mumbai Indians v Rising Pune Supergiant', 'Rising Pune Supergiant v Kolkata Knight Riders', 'Royal Challengers Bangalore v Gujarat Lions', 'Rising Pune

Supergiant v Royal Challengers Bangalore', 'Gujarat Lions v Mumbai Indians', 'Rising Pune Supergiant v Gujarat Lions', 'Kolkata Knight Riders v Rising Pune Supergiant', 'Delhi Daredevils v Gujarat Lions', 'Sunrisers Hyderabad v Rising Pune Supergiant', 'Kings XI Punjab v Gujarat Lions', 'Gujarat Lions v Delhi Daredevils', 'Delhi Daredevils v Rising Pune Supergiant', 'Gujarat Lions v Sunrisers Hyderabad', 'Rising Pune Supergiant v Kings XI Punjab', 'Mumbai Indians v Rising Pune Supergiants', 'Gujarat Lions v Rising Pune Supergiants', 'Kings XI Punjab v Rising Pune Supergiants', 'Rising Pune Supergiants v Royal Challengers Bangalore', 'Rising Pune Supergiants v Kolkata Knight Riders', 'Sunrisers Hyderabad v Rising Pune Supergiants', 'Rising Pune Supergiants v Gujarat Lions', 'Rising Pune Supergiants v Mumbai Indians', 'Delhi Daredevils v Rising Pune Supergiants', 'Royal Challengers Bangalore v Rising Pune Supergiants', 'Rising Pune Supergiants v Sunrisers Hyderabad', 'Kolkata Knight Riders v Rising Pune Supergiants', 'Rising Pune Supergiants v Delhi Daredevils', 'Rising Pune Supergiants v Kings XI Punjab', 'Sunrisers Hyderabad v Pune Warriors', 'Pune Warriors v Kings XI Punjab', 'Pune Warriors v Rajasthan Royals', 'Mumbai Indians v Pune Warriors', 'Chennai Super Kings v Pune Warriors', 'Pune Warriors v Sunrisers Hyderabad', 'Kings XI Punjab v Pune Warriors', 'Royal Challengers Bangalore v Pune Warriors', 'Delhi Daredevils v Pune Warriors', 'Pune Warriors v Chennai Super Kings', 'Pune Warriors v Royal Challengers Bangalore', 'Rajasthan Royals v Pune Warriors', 'Pune Warriors v Kolkata Knight Riders', 'Pune Warriors v Mumbai Indians', 'Kolkata Knight Riders v Pune Warriors', 'Pune Warriors v Delhi Daredevils', 'Deccan Chargers v Chennai Super Kings', 'Deccan Chargers v Mumbai Indians', 'Rajasthan Royals v Deccan Chargers', 'Delhi Daredevils v Deccan Chargers', 'Deccan Chargers v Kolkata Knight Riders', 'Kolkata Knight Riders v Deccan Chargers', 'Pune Warriors v Deccan Chargers', 'Mumbai Indians v Deccan Chargers', 'Deccan Chargers v Pune Warriors', 'Chennai Super Kings v Deccan Chargers', 'Royal Challengers Bangalore v Deccan Chargers', 'Deccan Chargers v Kings XI Punjab', 'Deccan Chargers v Delhi Daredevils', 'Kings XI Punjab v Deccan Chargers', 'Deccan Chargers v Rajasthan Royals', 'Deccan Chargers v Royal Challengers Bangalore', 'Kochi Tuskers Kerala v Royal Challengers Bangalore', 'Pune Warriors v Kochi Tuskers Kerala', 'Mumbai Indians v Kochi Tuskers Kerala', 'Kochi Tuskers Kerala v Chennai Super Kings', 'Kolkata Knight Riders v Kochi Tuskers Kerala', 'Rajasthan Royals v Kochi Tuskers Kerala', 'Kochi Tuskers Kerala v Deccan Chargers', 'Kochi Tuskers Kerala v Delhi Daredevils', 'Delhi Daredevils v Kochi Tuskers Kerala', 'Kochi Tuskers Kerala v Kolkata Knight Riders', 'Royal Challengers Bangalore v Kochi Tuskers Kerala', 'Kochi Tuskers Kerala v Kings XI Punjab', 'Kochi Tuskers Kerala v Rajasthan Royals', 'Chennai Super Kings v Kochi Tuskers Kerala']

```
name_df = df[["name"]]
print(name_df.head())
```

```

                                name
0      Chennai Super Kings v Gujarat Titans
1      Lucknow Super Giants v Mumbai Indians
2                                TBC v TBC
3                                TBC v TBC
4      Royal Challengers Bangalore v Gujarat Titans

df[["name"]].to_csv("name_list.csv", index=False)

print(df.columns.tolist())

['season', 'id', 'name', 'short_name', 'description', 'home_team',
'away_team', 'toss_won', 'decision', '1st_inning_score',
'2nd_inning_score', 'winner', 'result', 'start_date', 'end_date',
'venue_id', 'venue_name', 'home_captain', 'away_captain', 'pom',
'points', 'super_over', 'home_overs', 'home_runs', 'home_wickets',
'home_boundaries', 'away_overs', 'away_runs', 'away_wickets',
'away_boundaries', 'highlights', 'home_key_batsman',
'home_key_bowler', 'home_playx1', 'away_playx1', 'away_key_batsman',
'away_key_bowler', 'match_days', 'umpire1', 'umpire2', 'tv_umpire',
'referee', 'reserve_umpire']

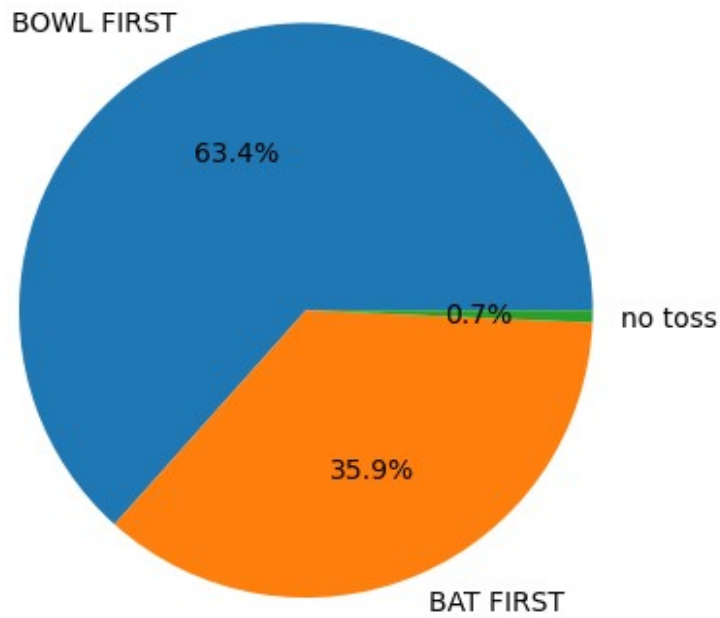
import matplotlib.pyplot as plt

df["decision"].value_counts().plot(
    kind="pie",
    autopct="%1.1f%%"
)

plt.title("Toss Decision Distribution")
plt.ylabel("")
plt.show()

```

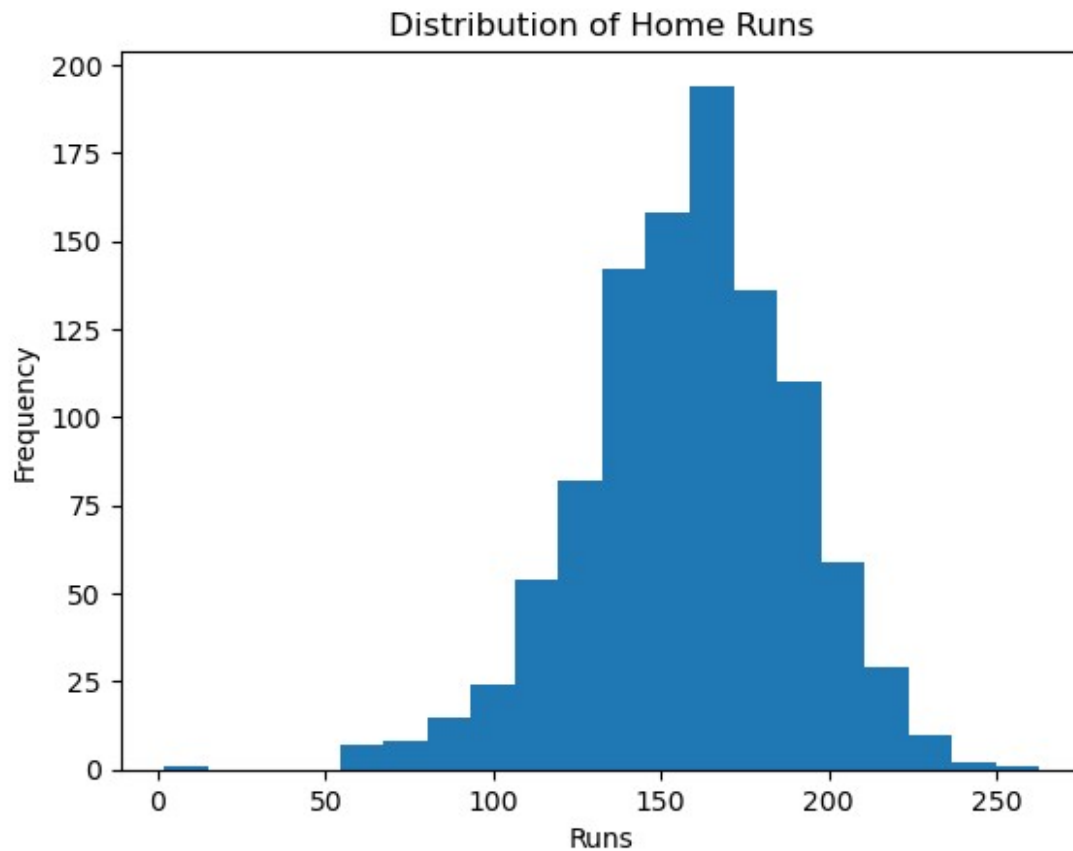
Toss Decision Distribution



```
import matplotlib.pyplot as plt

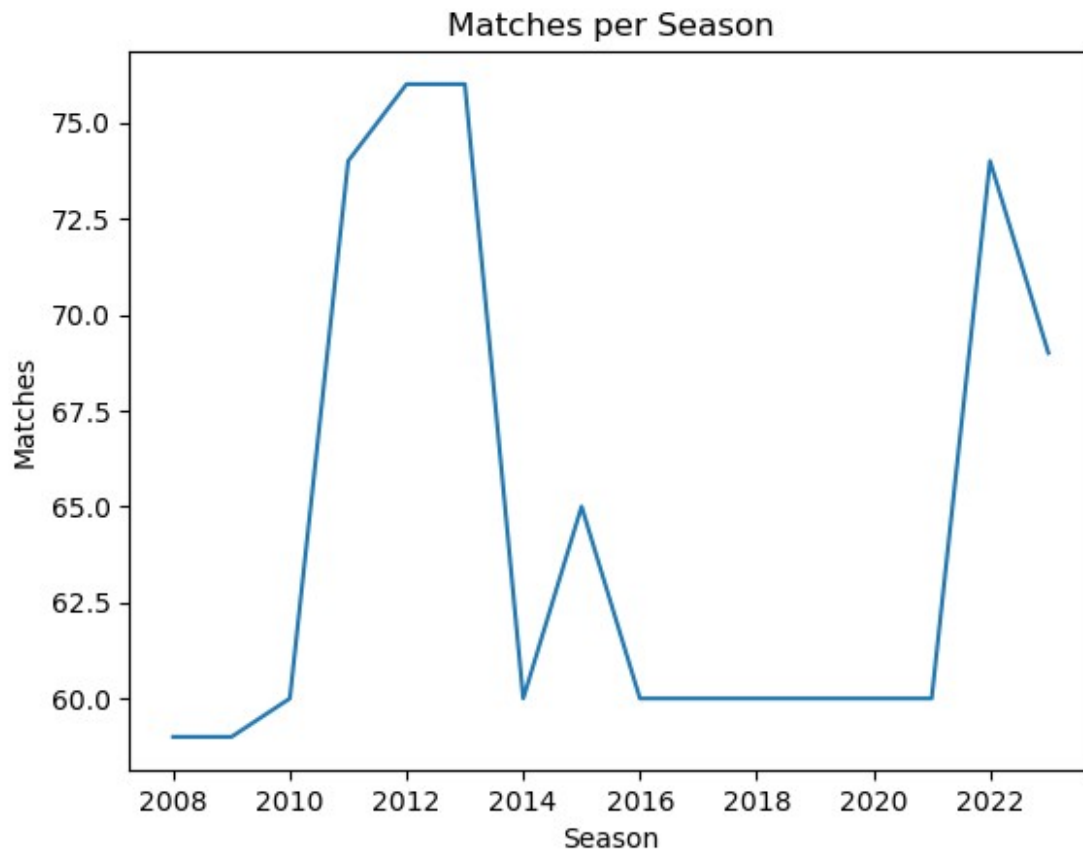
plt.hist(df["home_runs"].dropna(), bins=20)

plt.title("Distribution of Home Runs")
plt.xlabel("Runs")
plt.ylabel("Frequency")
plt.show()
```



```
matches_per_season = df["season"].value_counts().sort_index()
plt.plot(matches_per_season.index, matches_per_season.values)

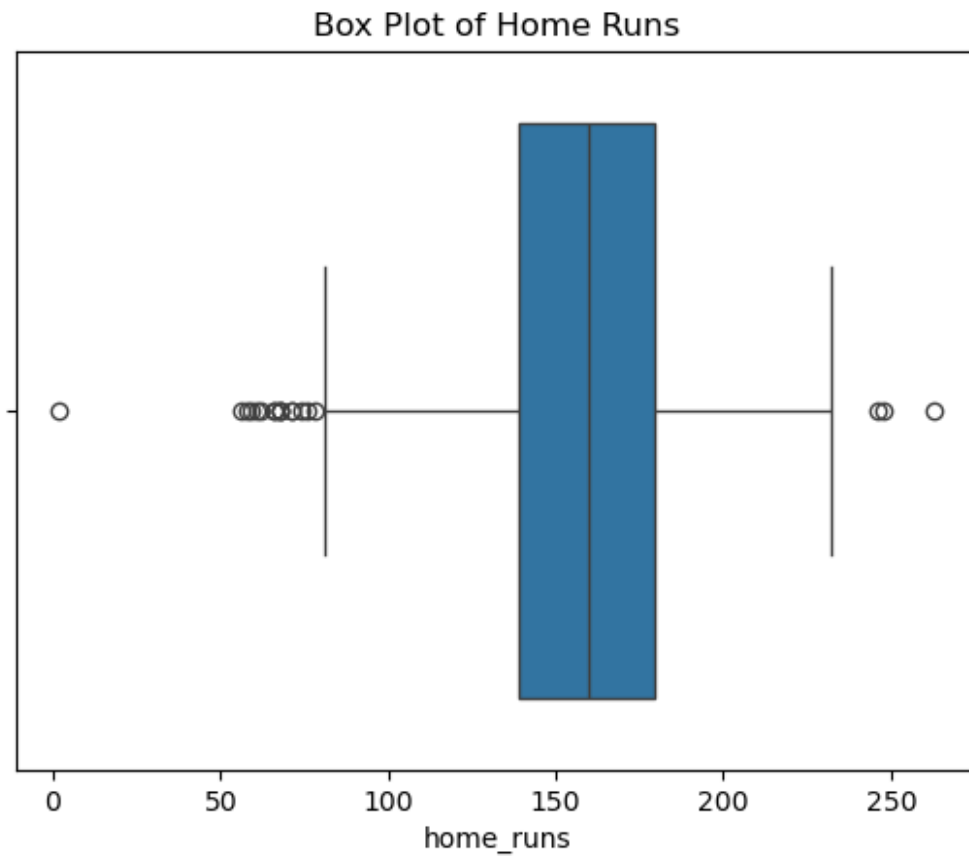
plt.title("Matches per Season")
plt.xlabel("Season")
plt.ylabel("Matches")
plt.show()
```

```
import seaborn as sns
import matplotlib.pyplot as plt

sns.boxplot(x=df["home_runs"])

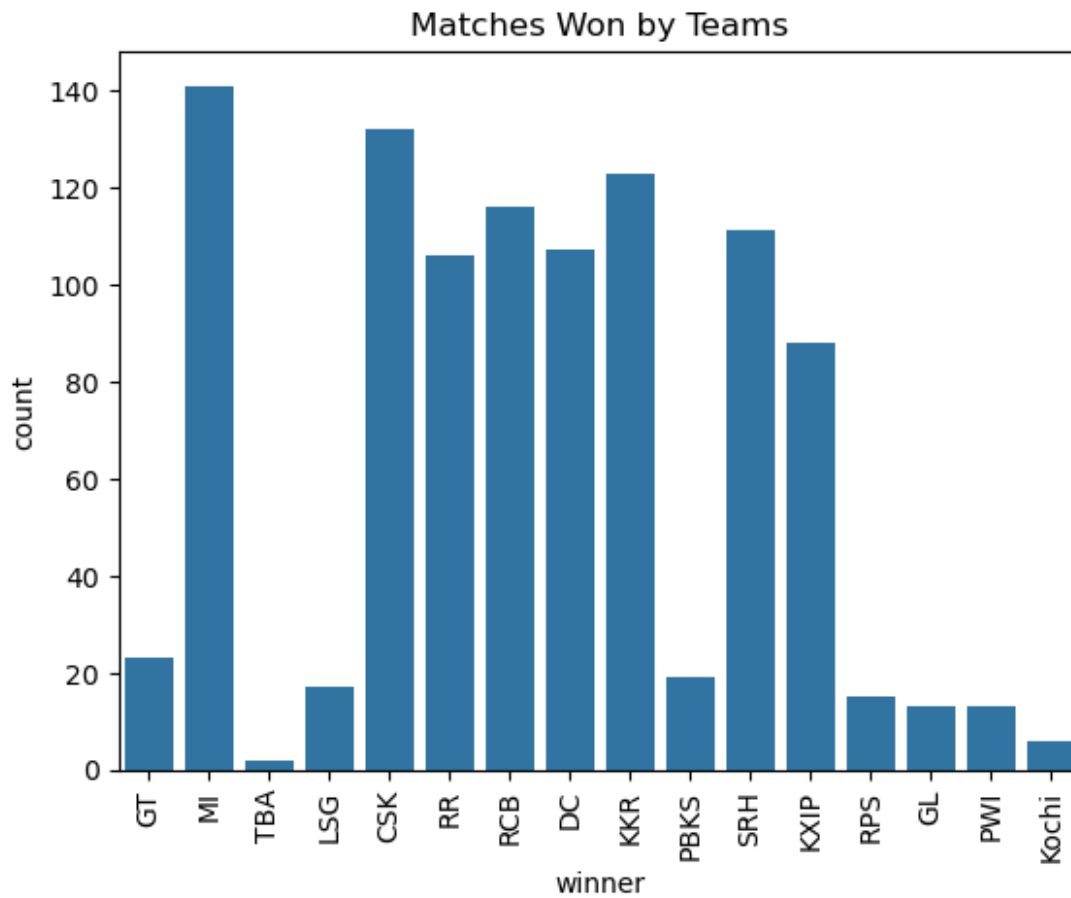
plt.title("Box Plot of Home Runs")
plt.show()
```



```
import seaborn as sns
import matplotlib.pyplot as plt

sns.countplot(data=df, x="winner")

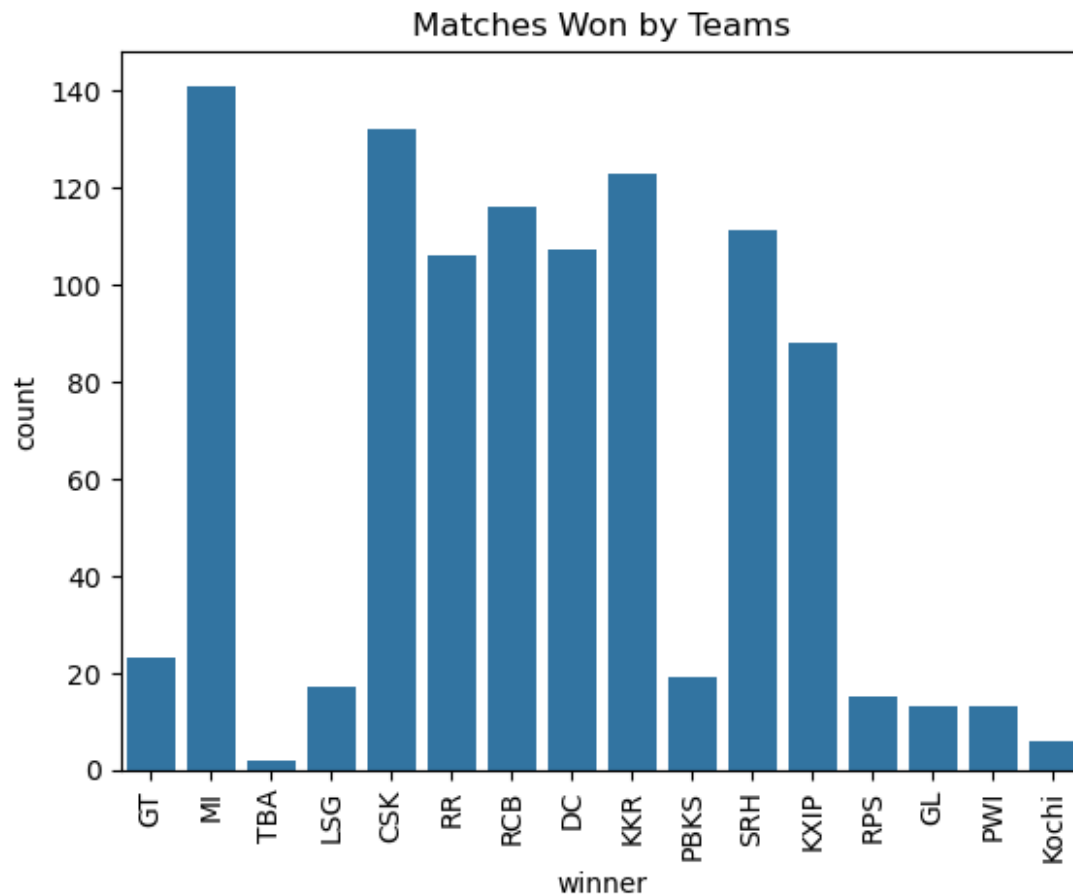
plt.xticks(rotation=90)
plt.title("Matches Won by Teams")
plt.show()
```



```
import seaborn as sns
import matplotlib.pyplot as plt

sns.countplot(data=df, x="winner")

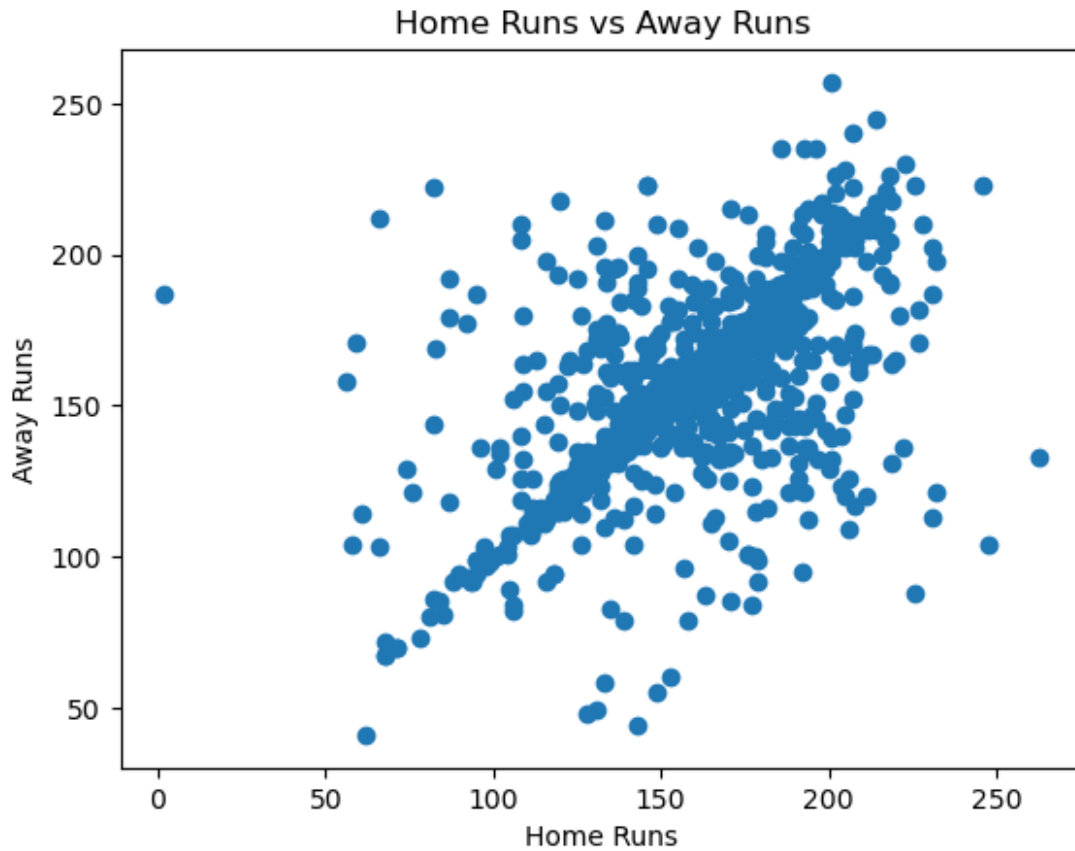
plt.xticks(rotation=90)
plt.title("Matches Won by Teams")
plt.show()
```



```
import matplotlib.pyplot as plt

plt.scatter(df["home_runs"], df["away_runs"])

plt.title("Home Runs vs Away Runs")
plt.xlabel("Home Runs")
plt.ylabel("Away Runs")
plt.show()
```



```
import seaborn as sns
import matplotlib.pyplot as plt

corr = df.select_dtypes(include="number").corr()

plt.figure(figsize=(10,8))
sns.heatmap(corr, annot=True, cmap="coolwarm")

plt.title("Correlation Heatmap")
plt.show()
```

